

An Introduction to the quantum simulation

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- Richard Feynman and Yuri Manin independently proposed the idea of using quantum simulators to simulate quantum systems.



Richard P. Feynman



Yuri I. Manin (Юрий Манин)

Yu. I. Manin, “Computable and Non-Computable,”
Sovetskoe Radio, Moscow, 1980, 128 p.

R.P. Feynman, Simulating physics with computers,
Internat. J. Theoret. Phys. 21 (6–7) (1982) 467–488,
<http://dx.doi.org/10.1007/bf02650179>.

What is quantum simulation?

- Quantum simulation involves using a controllable quantum system to study the behaviour of another quantum system that is difficult to study directly.
- Richard Feynman and Yuri Manin independently proposed the idea of using quantum simulators to simulate quantum systems.
- This idea of using quantum systems to simulate other quantum systems inspired the development of quantum computing.

Why do we need quantum simulation?

- Unlike classical simulations, quantum simulations leverage the principles of quantum mechanics to model complex quantum phenomena that are infeasible for classical computers.

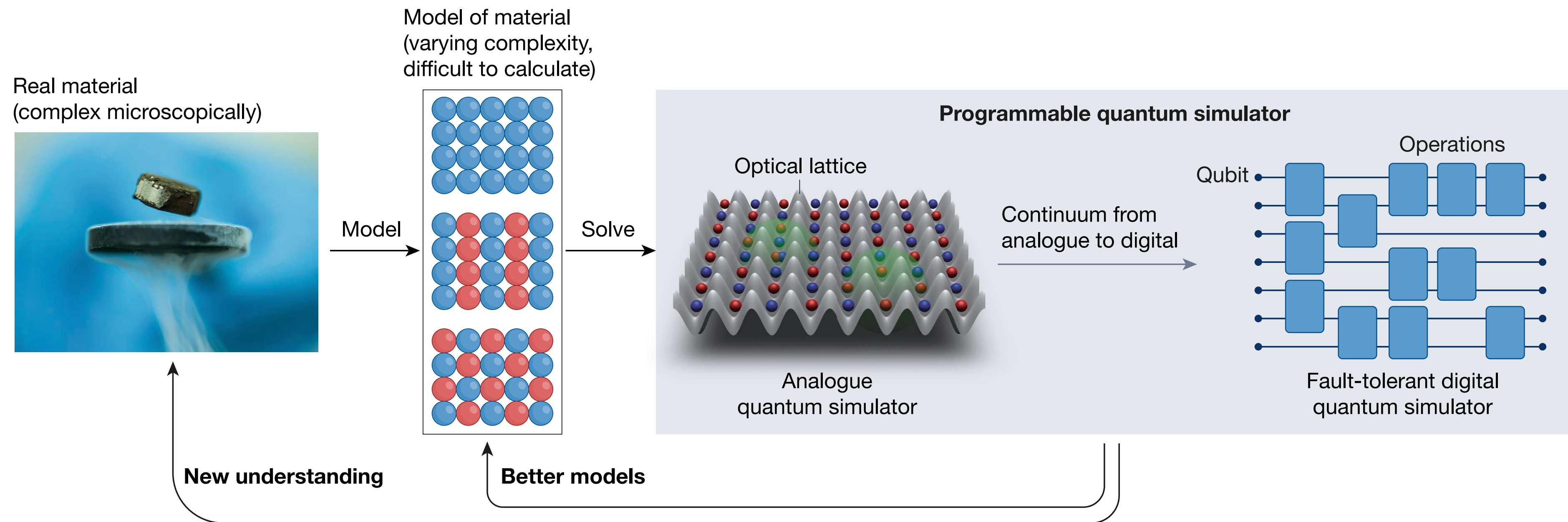
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- Classical computers struggle with simulating quantum systems due to the exponential growth of the Hilbert space.

Why do we need quantum simulation?

- Unlike classical simulations, quantum simulations leverage the principles of quantum mechanics to model complex quantum phenomena that are infeasible for classical computers.
- Classical computers struggle with simulating quantum systems due to the exponential growth of the Hilbert space.
- Quantum simulations can provide insights into many-body physics, quantum chemistry, and material science.

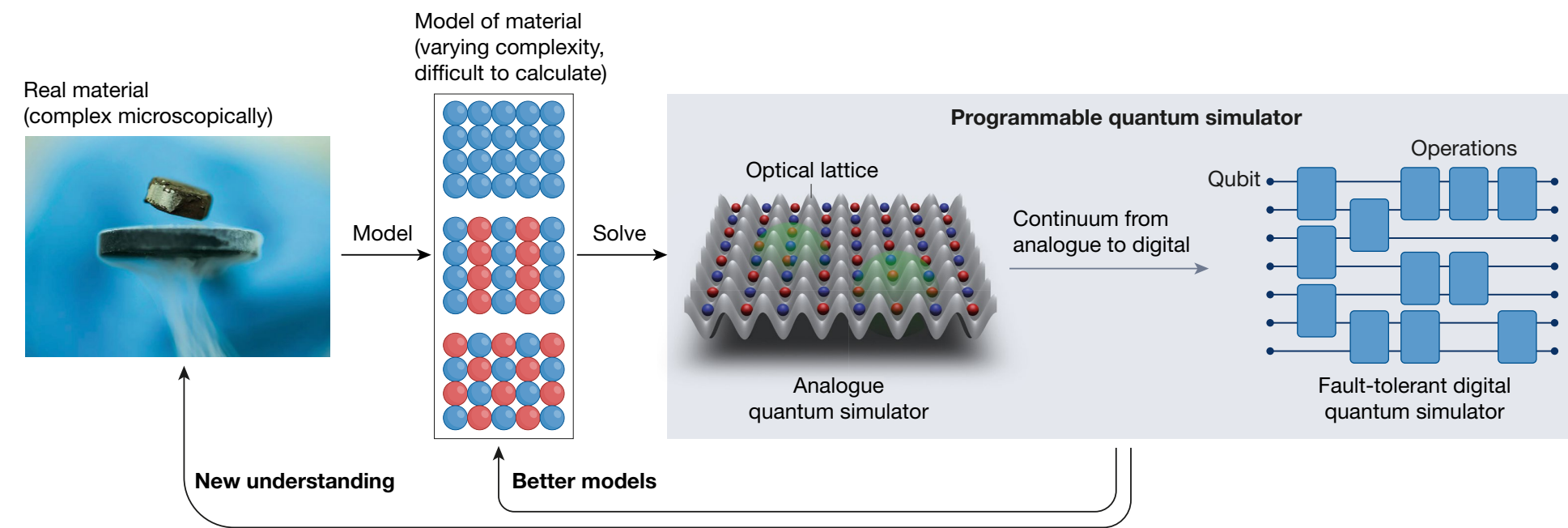
Quantum analogue vs digital simulations



Example platforms for analogue quantum simulation:

- Ultracold atoms in optical lattices.
- Trapped ions.
- Atom arrays with Rydberg interactions.
- Superconducting circuits.
- Photonic waveguides or beamsplitter arrays.

Quantum analogue vs digital simulations

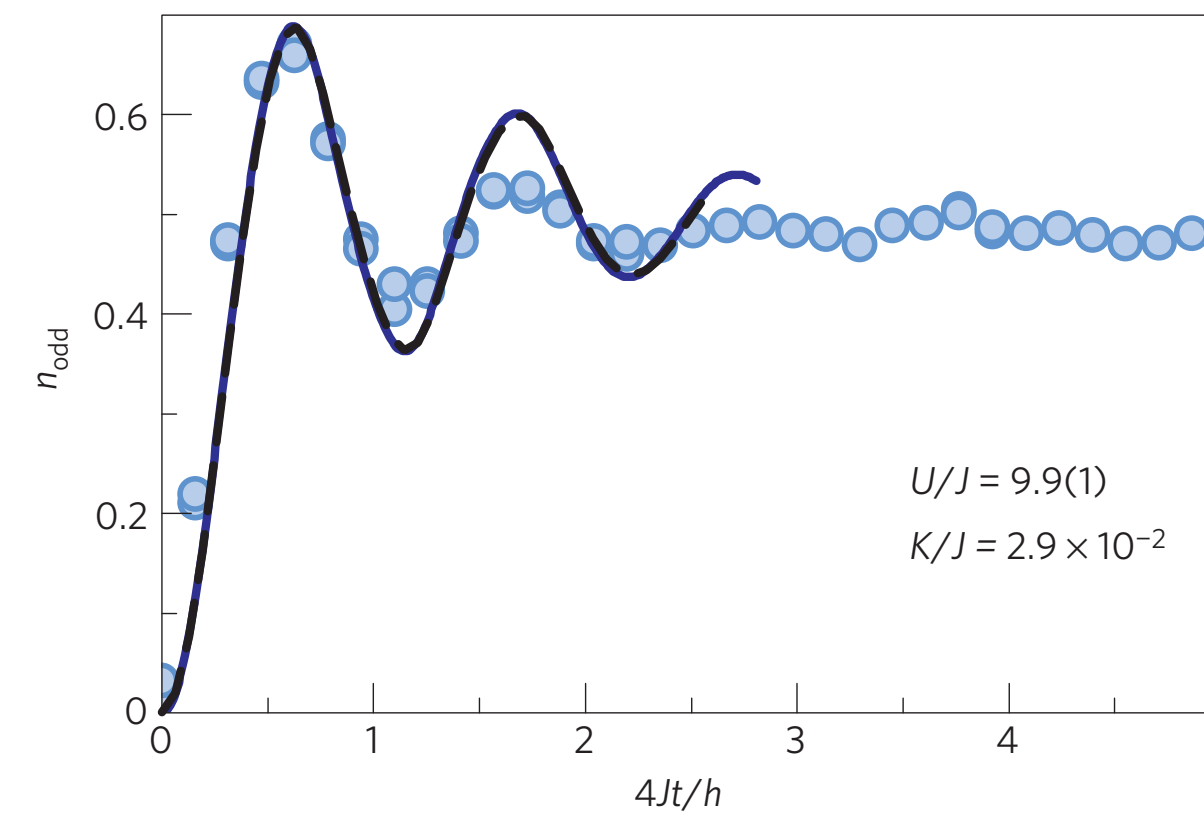
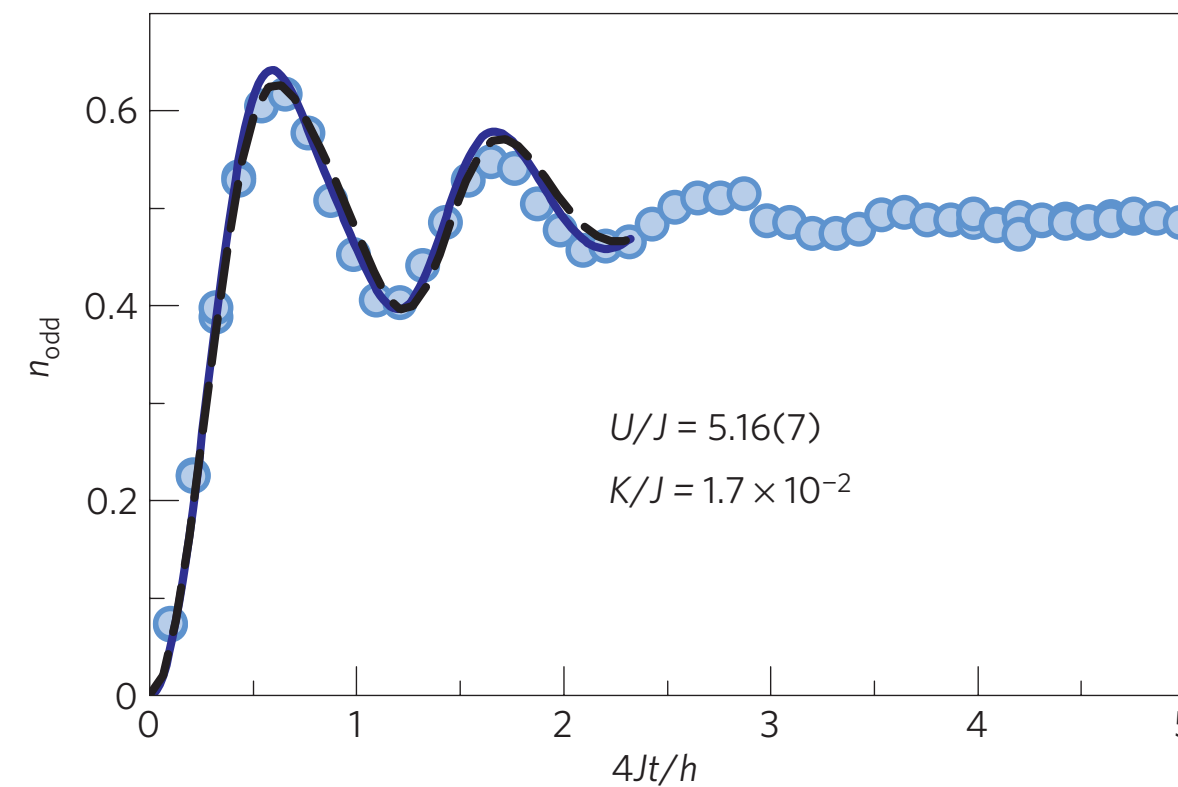


Hamiltonian describing a gas of interacting bosonic atoms in a periodic potential

$$H_{\text{sim}} = -J \sum_{i,j} \hat{a}_i^\dagger \hat{a}_j + \sum_i \varepsilon_i \hat{n}_i + \frac{1}{2} U \sum_i \hat{n}_i (\hat{n}_i - 1),$$

Bose-Hubbard Hamiltonian

$$H_{\text{BH}} = -J \sum_{i,j} \hat{b}_i^\dagger \hat{b}_j + \frac{1}{2} U \sum_i \hat{n}_i (\hat{n}_i - 1) - \mu \sum_i \hat{n}_i,$$



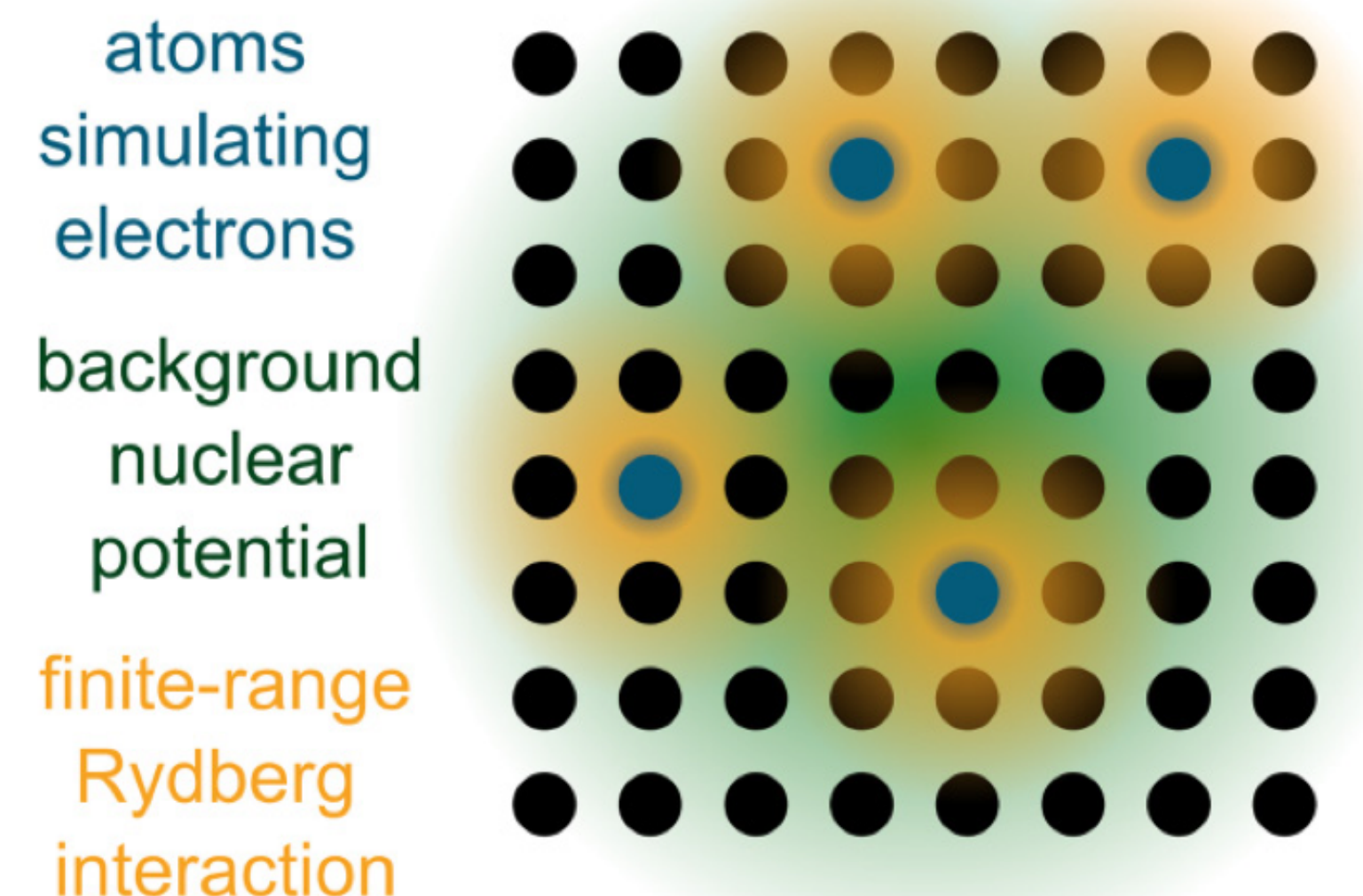
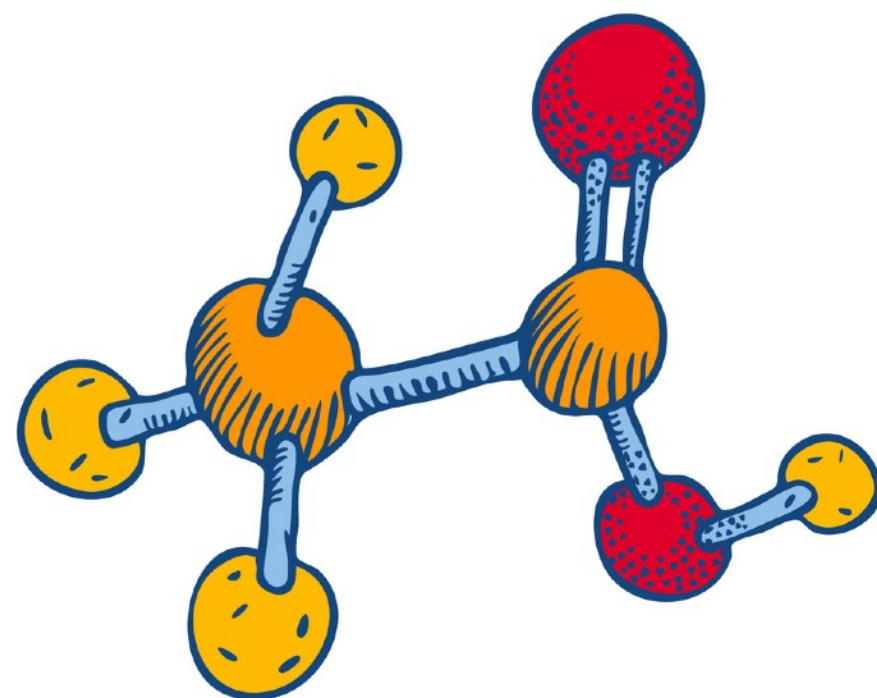
Hard to control errors!

Trotzky, S., Chen, YA., Flesch, A. *et al.* Probing the relaxation towards equilibrium in an isolated strongly correlated one-dimensional Bose gas. *Nature Phys* **8**, 325–330 (2012).

Types of quantum simulation

Analogue vs digital

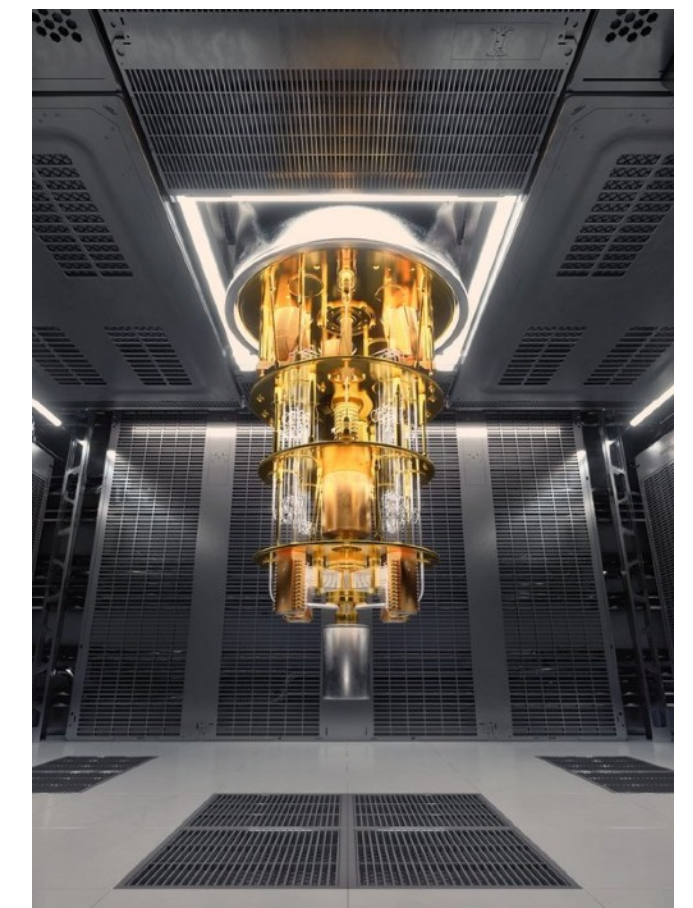
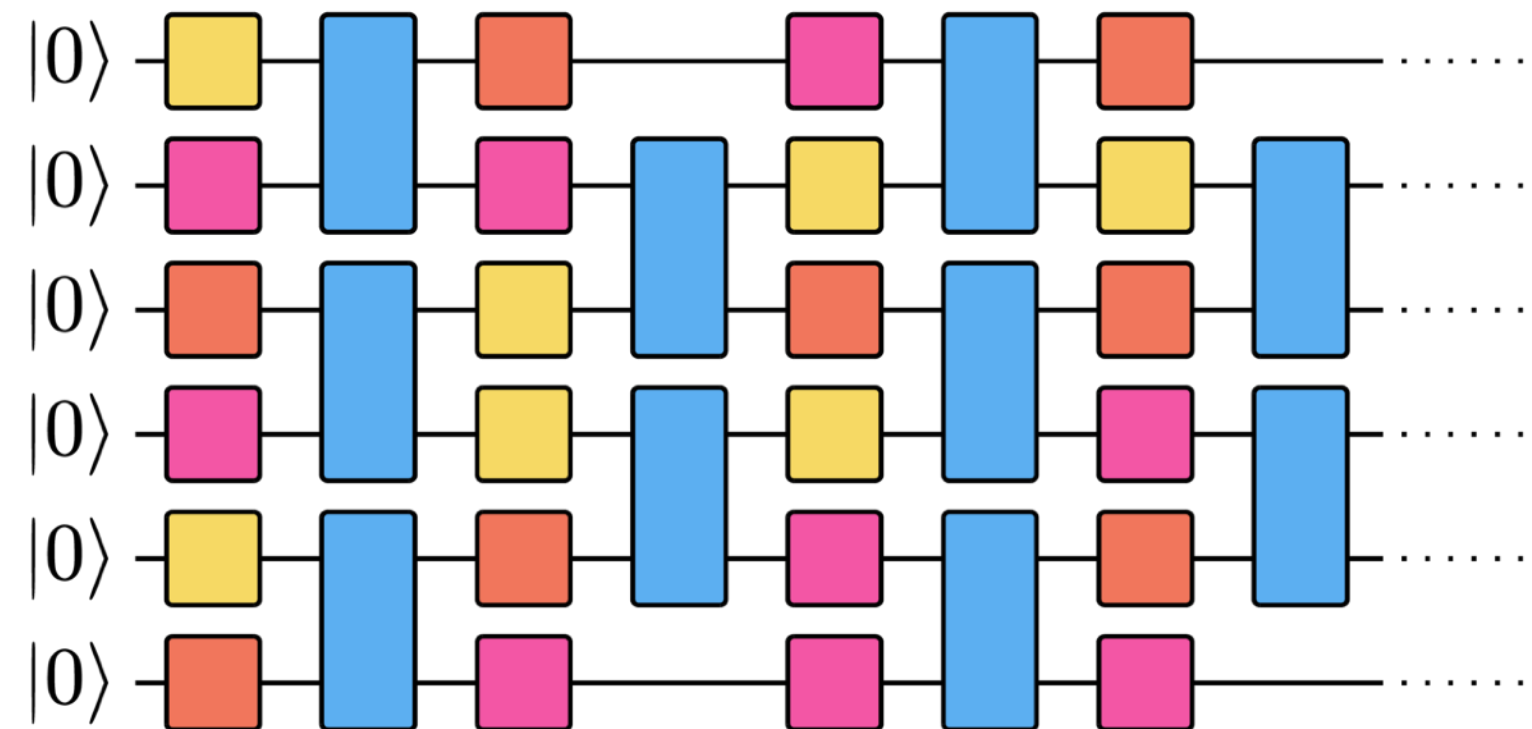
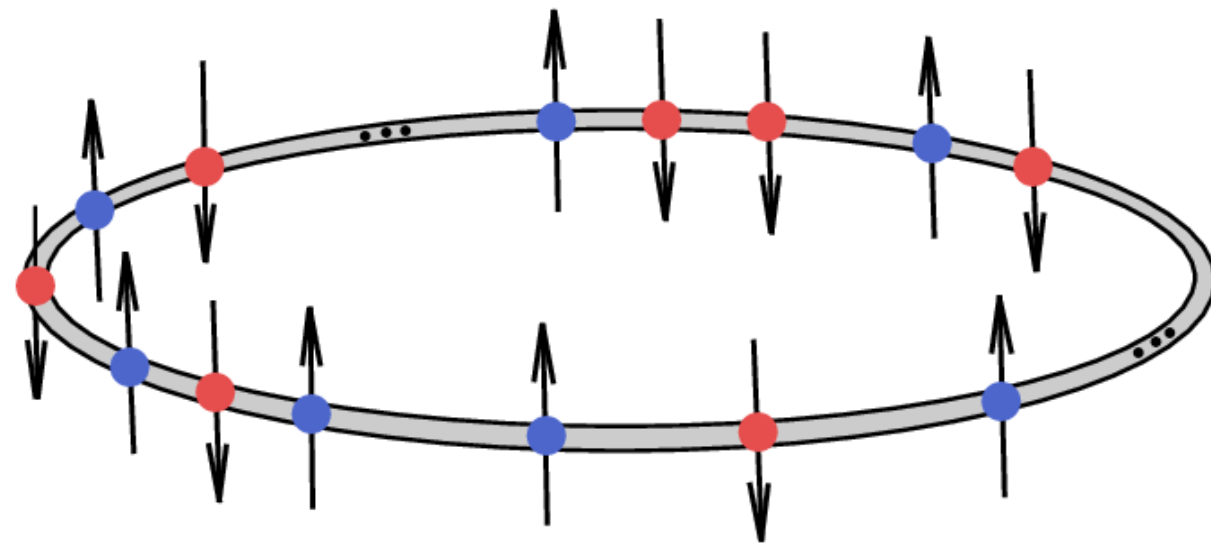
- **Analog Quantum Simulation:** Directly mimics the target quantum system using a physically similar system that is controllable.
 - ▶ This is done using analogue quantum computing which applies continuous transformations to continuous quantum states.
 - ▶ Example: Using fermionic atoms hopping in an optical lattice to simulate quantum chemistry (Malz D, Cirac JI. PRX Quantum. 2023 Apr 1;4(2):020301.)



Types of quantum simulation

Analogue vs digital

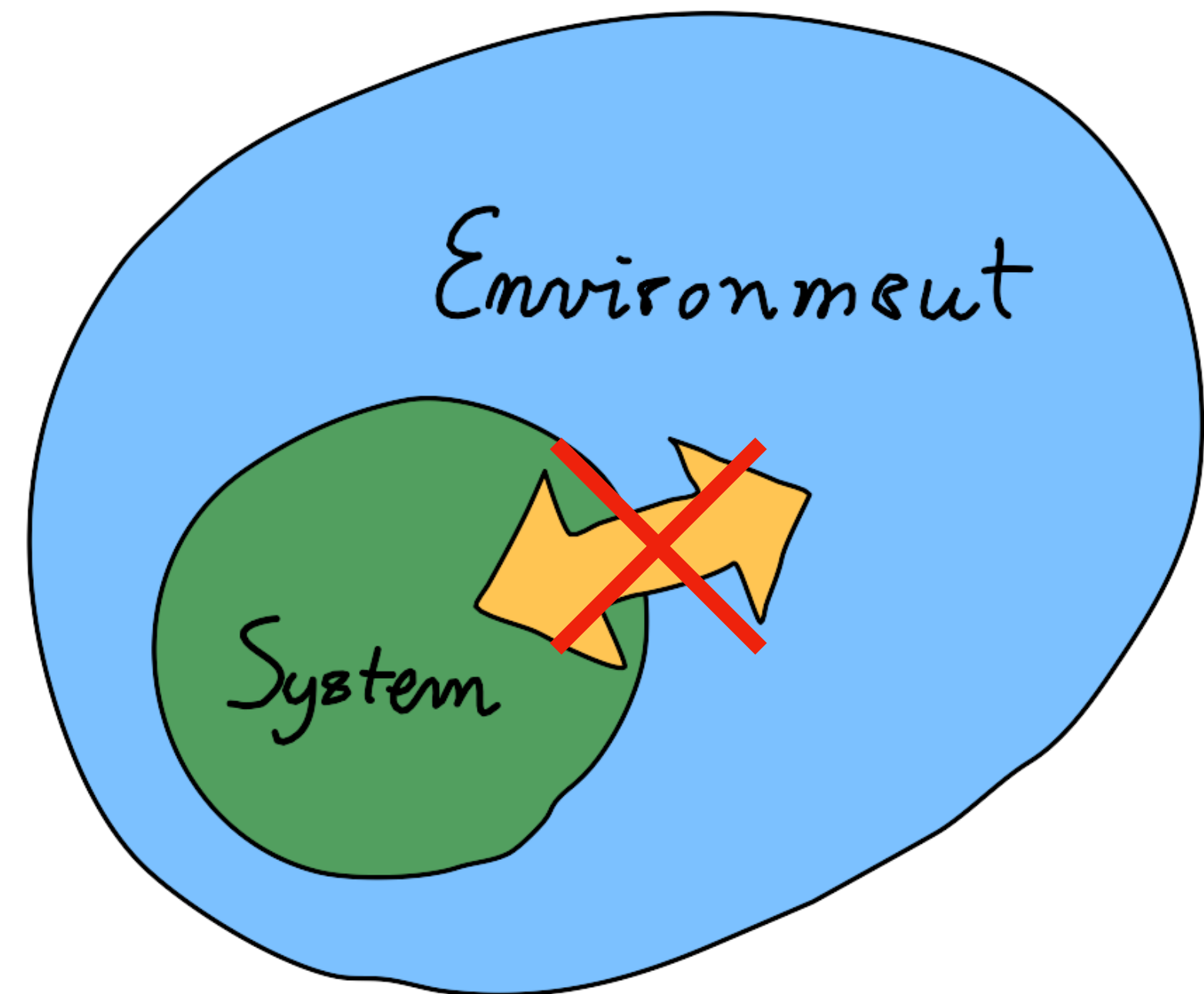
- **Digital Quantum Simulation:** Uses a quantum computer to simulate the quantum system by breaking down the simulation into a sequence of quantum gates.
 - ▶ This is called digital because it relies on digital quantum computing which breaks down the computation into a discrete set of operations.
 - ▶ Example: Using a quantum computer and a method such as Trotter-Suzuki product formulas to simulate Heisenberg spin chains.



What Types of quantum systems do we want to simulate?

What Types of quantum systems do we want to simulate?

- **Closed Quantum Systems:**
 - Isolated from their environment.
 - Evolve according to the Schrödinger equation with unitary time evolution.

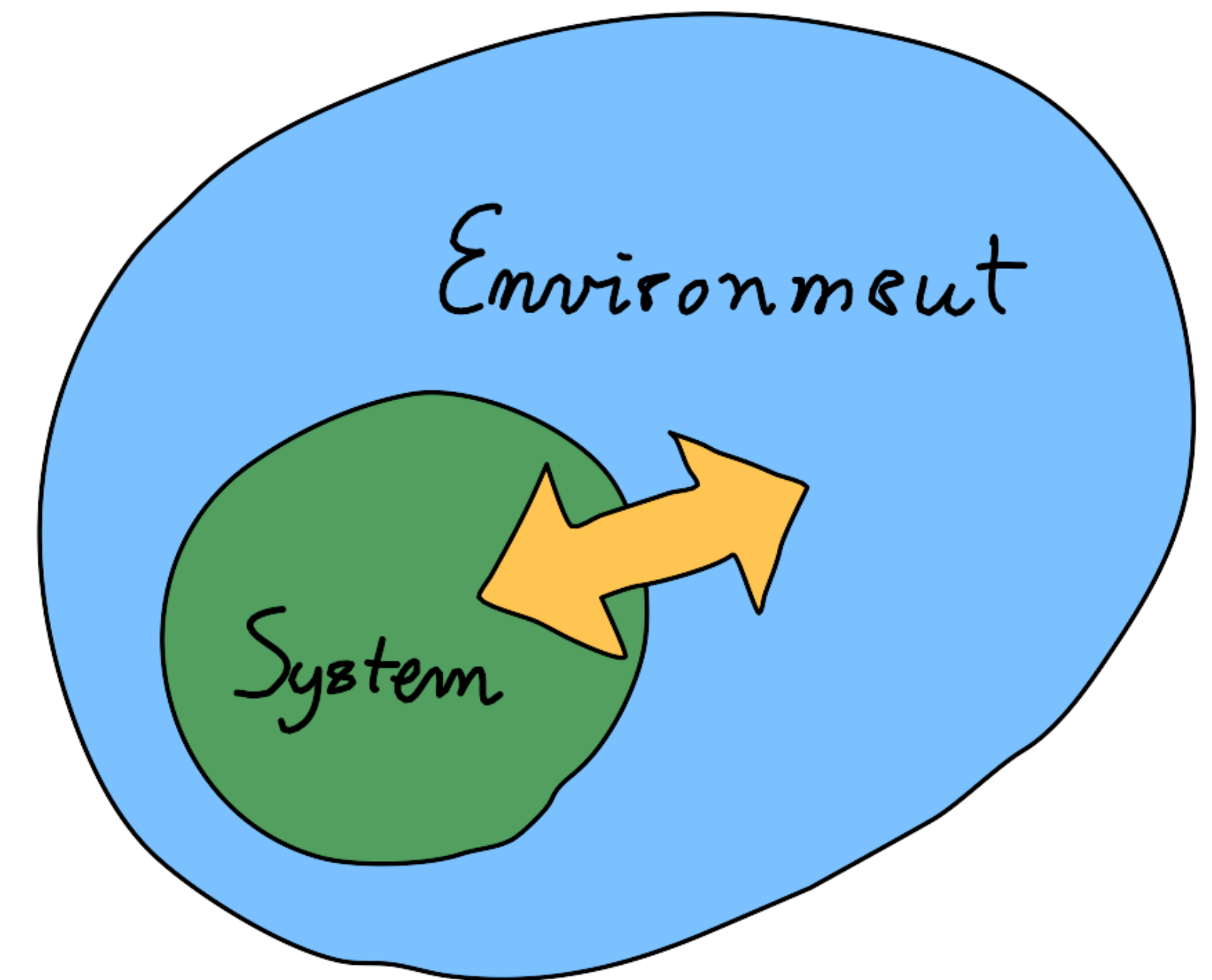


In this lecture

What Types of quantum systems do we want to simulate?

- **Open Quantum Systems:**

- ▶ The system is allowed to interact with an external environment so that it can exchange energy, matter and information.
- ▶ Evolve according to the Lindblad master equation, incorporating dissipation and decoherence.

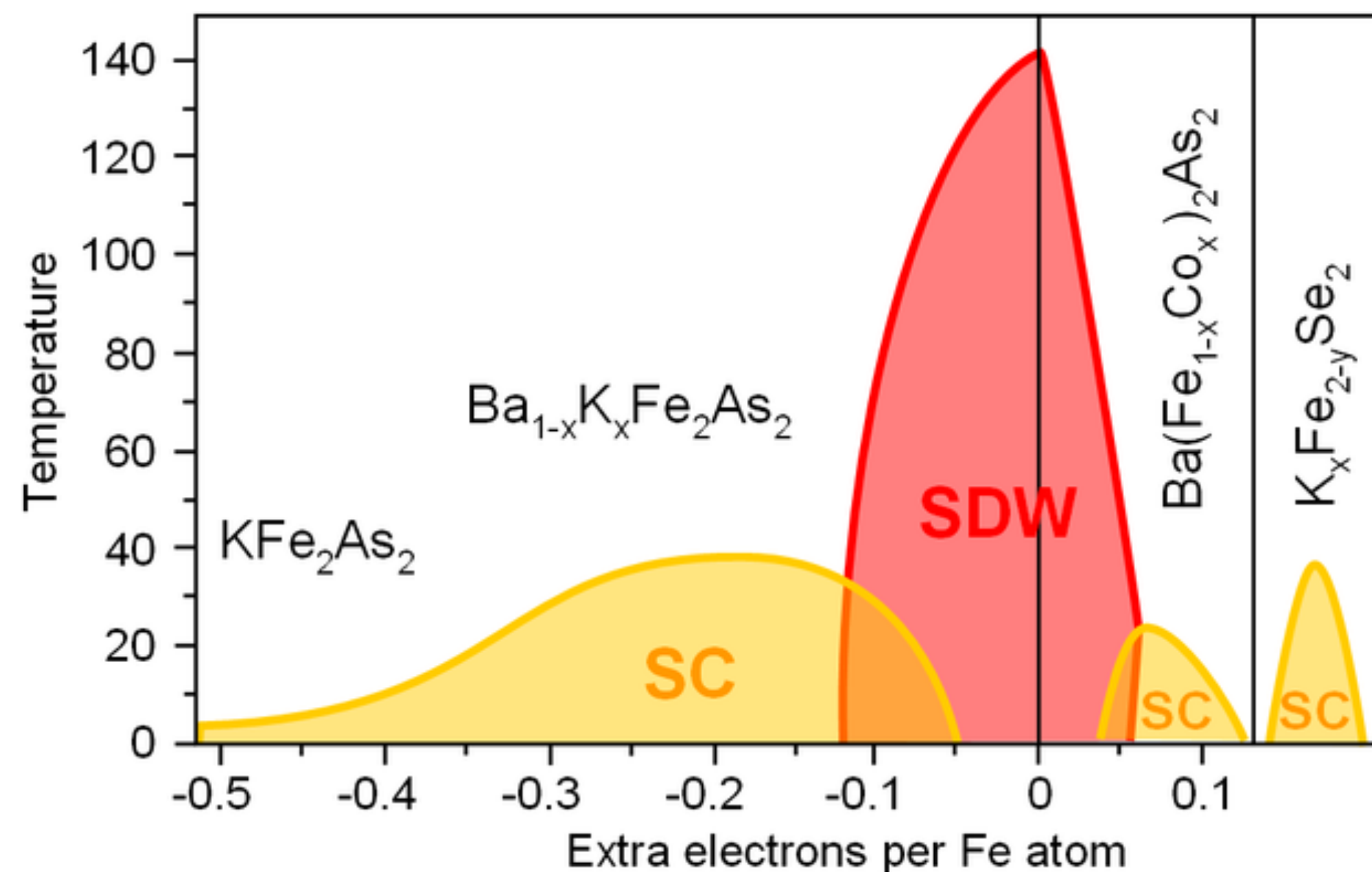


In my research

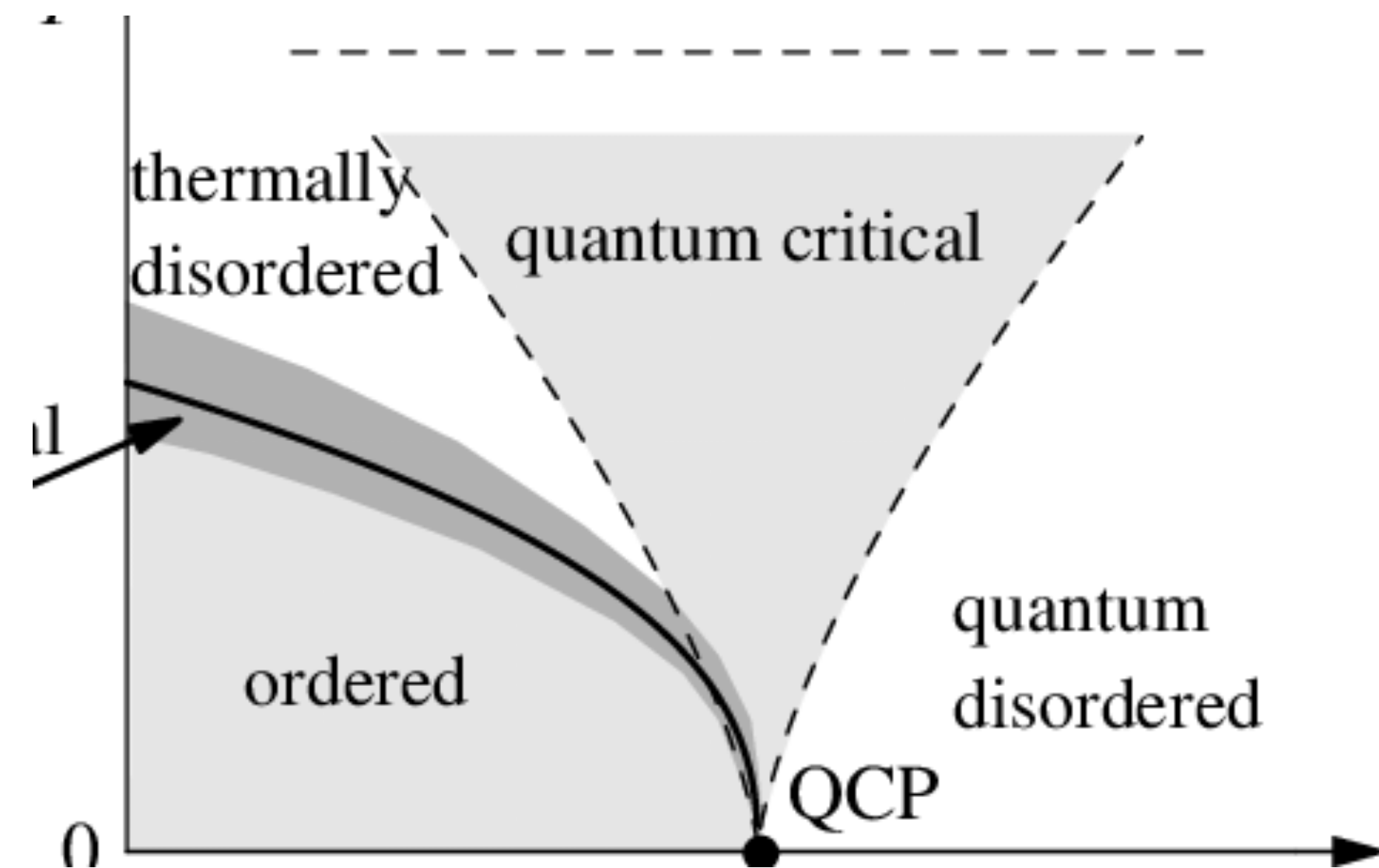
Applications of quantum simulation

- **Fundamental Physics:**

- Understanding high-temperature superconductivity.
- Studying quantum phase transitions.

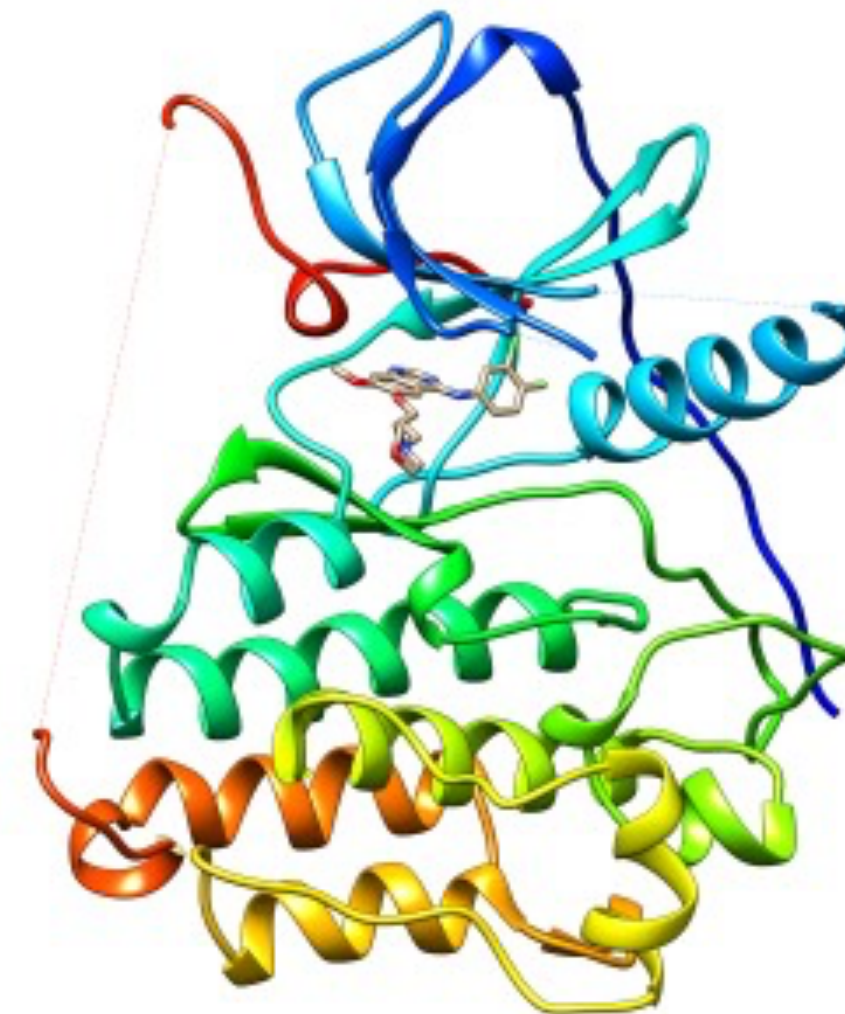
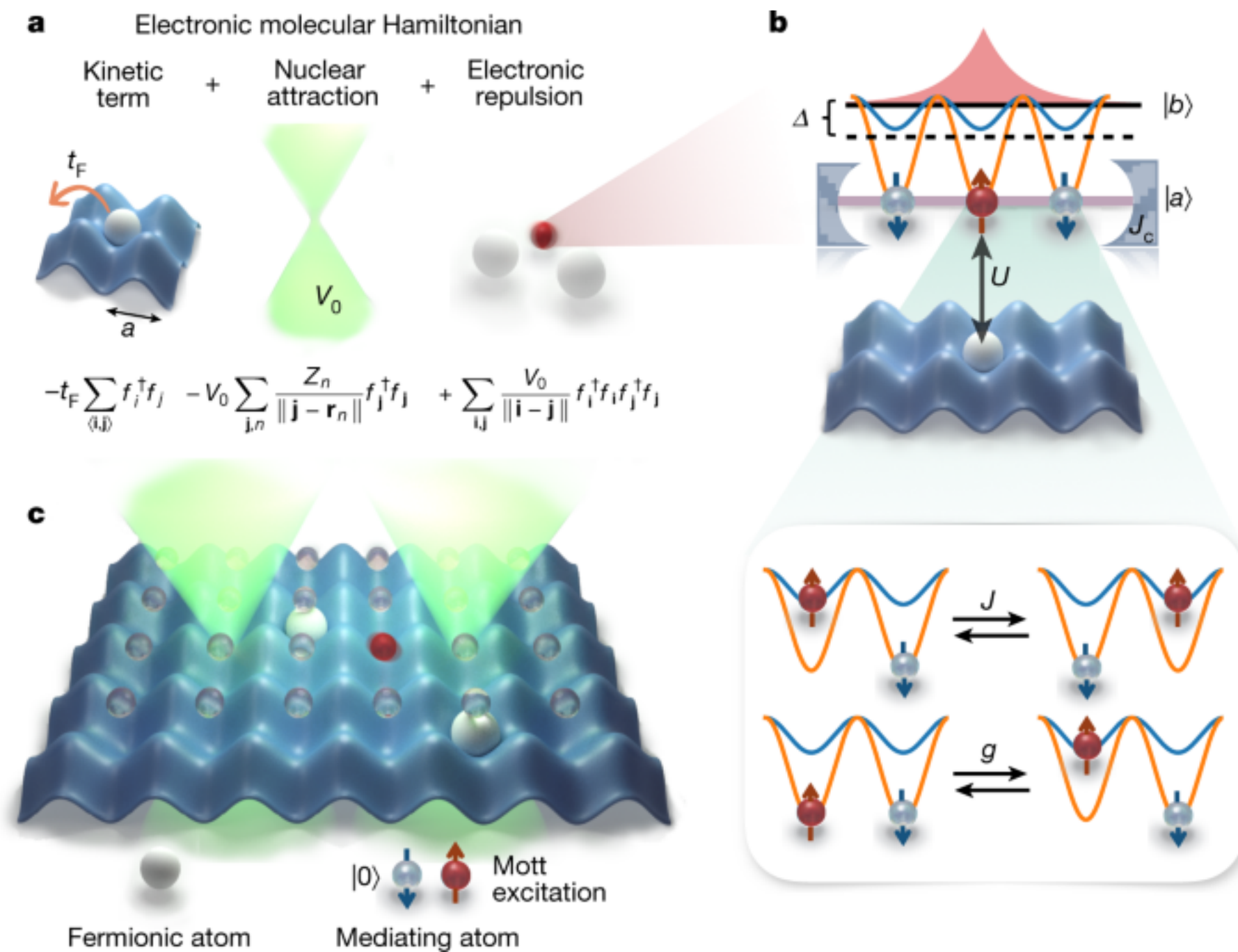


Phase diagram for high temperature super conductors based on iron



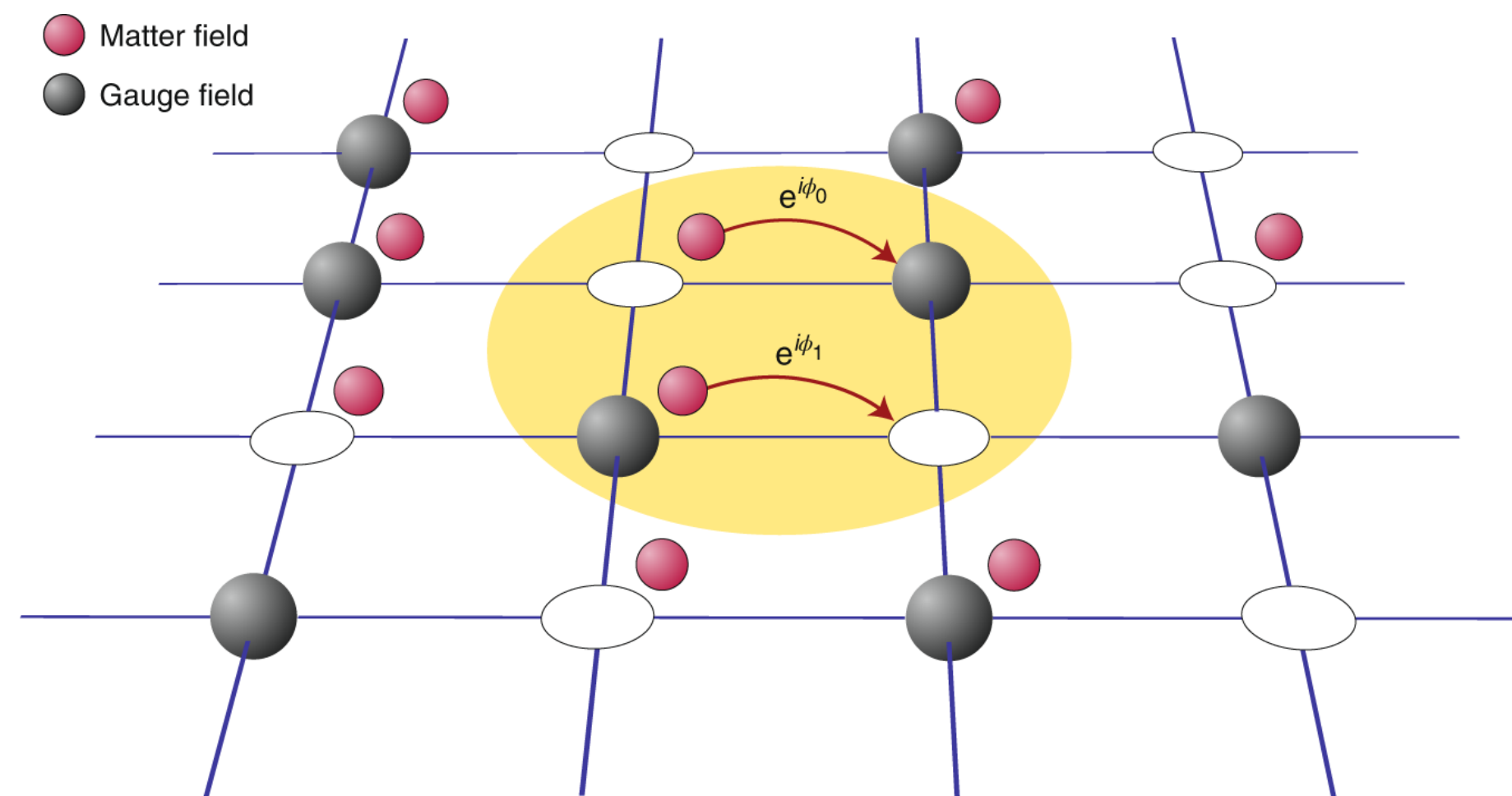
Applications of quantum simulation

- **Quantum Chemistry:**
 - Simulating molecular structures and reactions.
 - Designing new drugs and materials.



Applications of quantum simulation

- **High Energy Physics:**
 - Simulating lattice gauge theories and particle interactions.



Quantum Simulation today?

2019

2026

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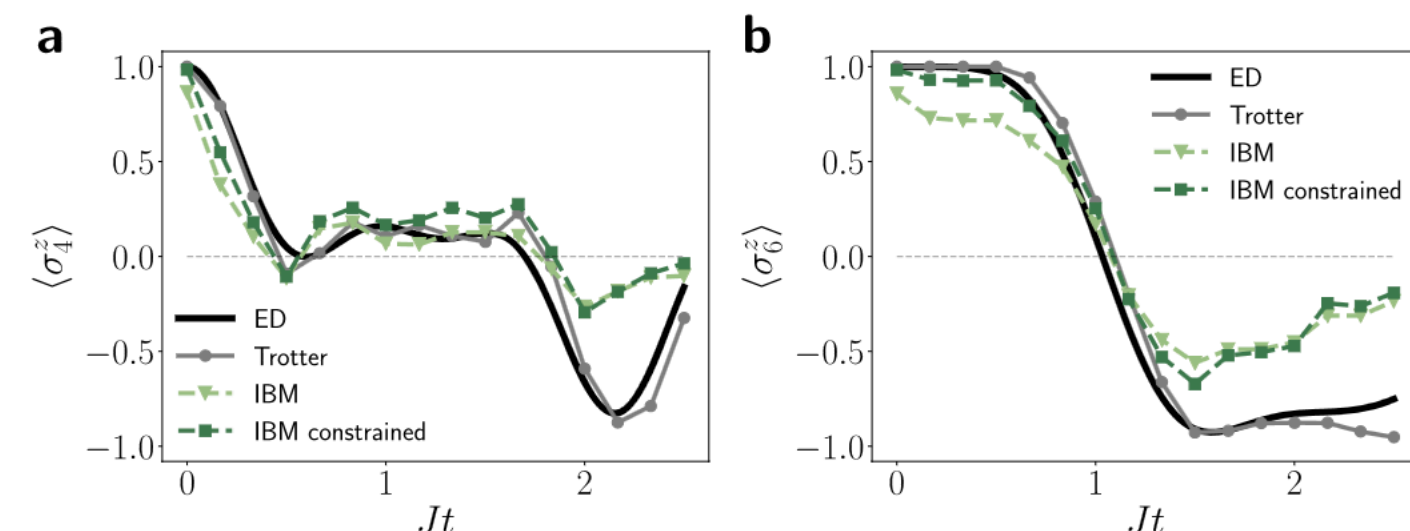
Simulating quantum many-body dynamics on a current digital quantum computer

Adam Smith^{1,2*}, M. S. Kim¹, Frank Pollmann² and Johannes Knolle¹

Universal quantum computers are potentially an ideal setting for simulating many-body quantum dynamics that is out of reach for classical digital computers. We use state-of-the-art IBM quantum computers to study paradigmatic examples of condensed matter physics—we simulate the effects of disorder and interactions on quantum particle transport, as well as correlation and entanglement spreading. Our benchmark results show that the quality of the current machines is below what is necessary for quantitatively accurate continuous-time dynamics of observables and reachable system sizes are small comparable to exact diagonalization. Despite this, we are successfully able to demonstrate clear qualitative behaviour associated with localization physics and many-body interaction effects.

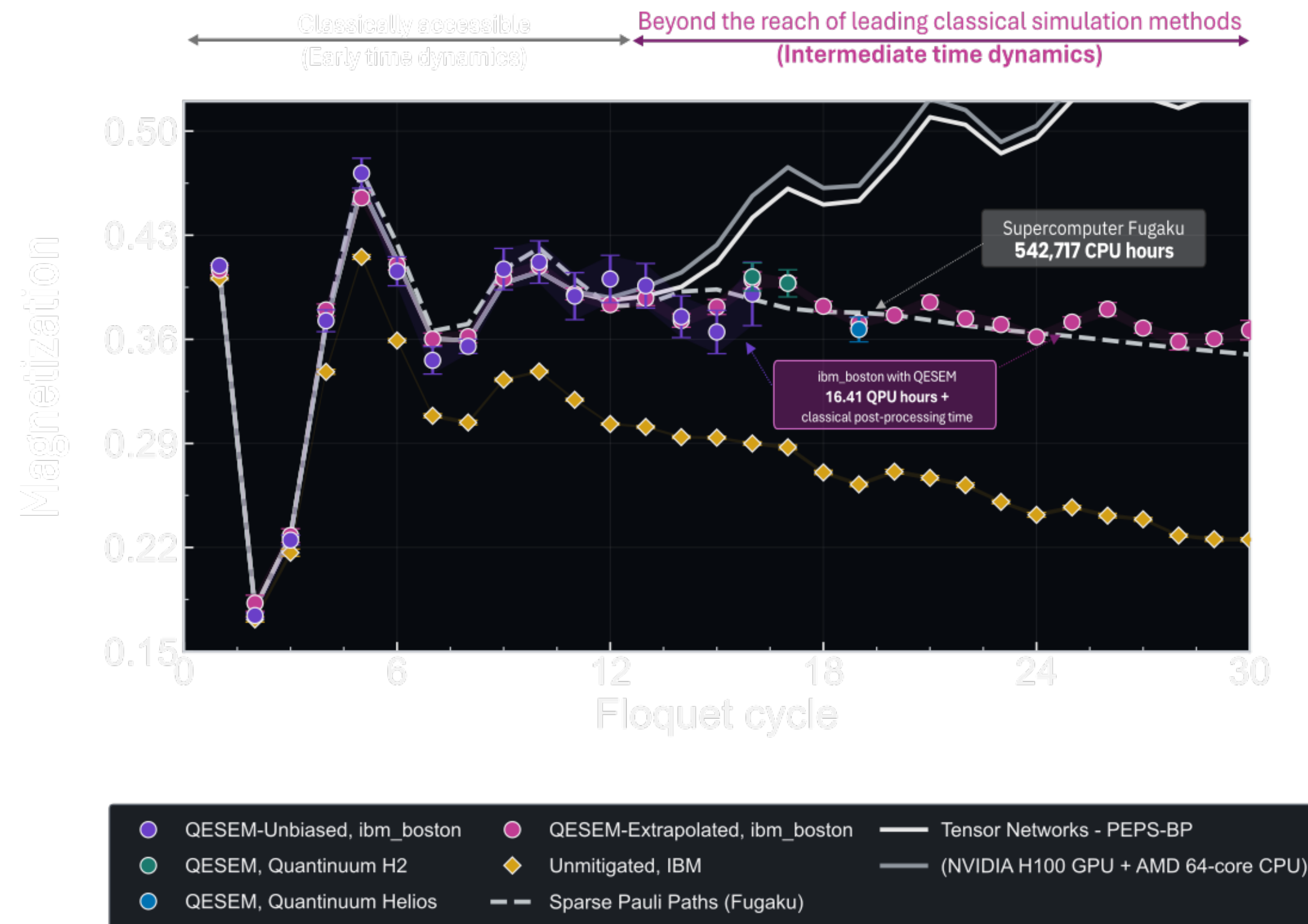
npj Quantum Information (2019)5:106; <https://doi.org/10.1038/s41534-019-0217-0>

$$\hat{H} = -J \sum_{j=1}^{N-1} (\hat{\sigma}_j^x \hat{\sigma}_{j+1}^x + \hat{\sigma}_j^y \hat{\sigma}_{j+1}^y) + U \sum_{j=1}^{N-1} \hat{\sigma}_j^z \hat{\sigma}_{j+1}^z + \sum_{j=1}^N h_j \hat{\sigma}_j^z,$$



“Our benchmark results show that the quality of the current machines is below what is necessary for quantitatively accurate continuous-time dynamics of observables and reachable system sizes are small comparable to exact diagonalization.”

QEDMA + IBM: arXiv:2607.24937



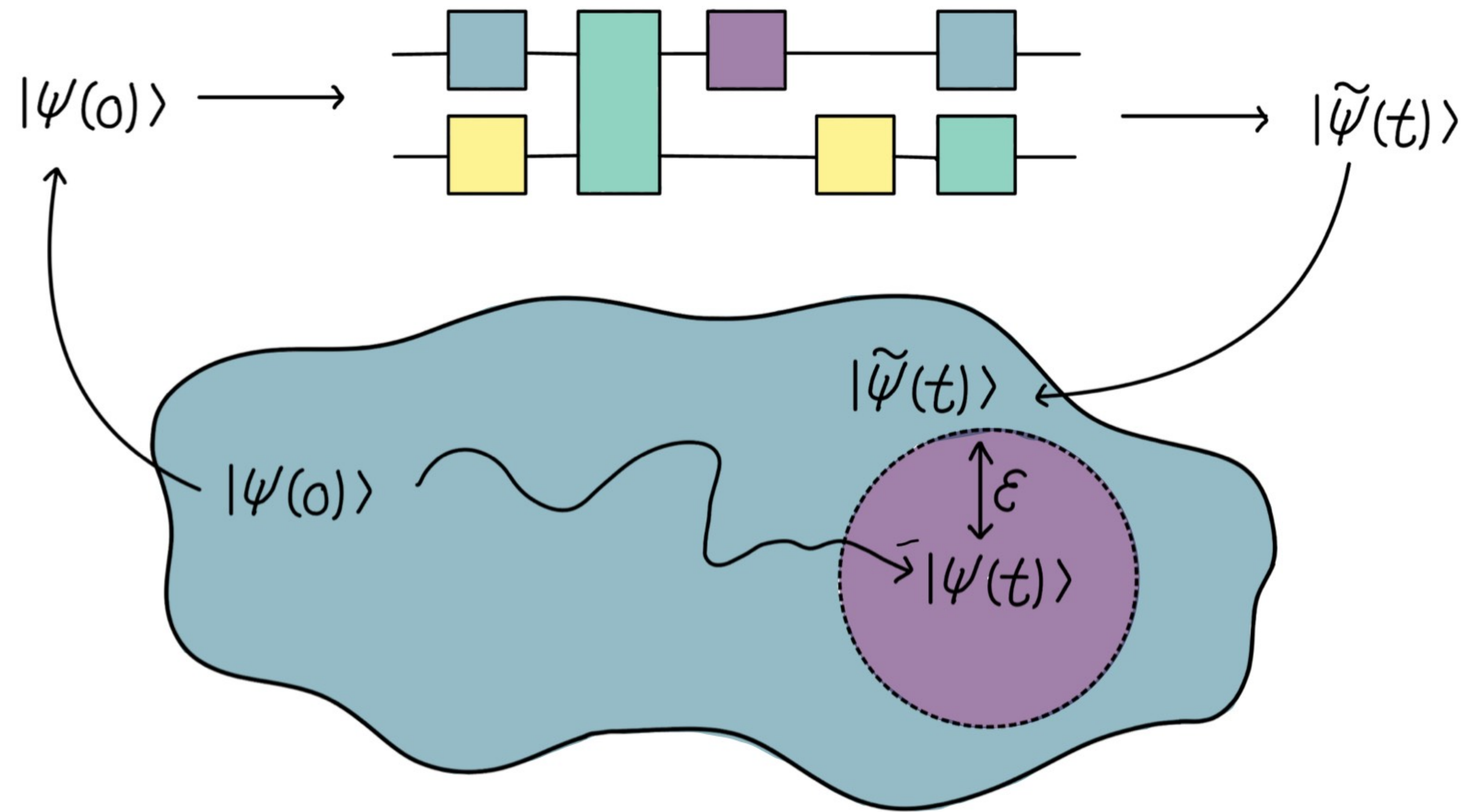
● QESEM-Unbiased, ibm_boston ● QESEM-Extrapolated, ibm_boston — Tensor Networks - PEPS-BP
 ● QESEM, Quantinuum H2 ◆ Unmitigated, IBM — (NVIDIA H100 GPU + AMD 64-core CPU)
 ● QESEM, Quantinuum Helios — Sparse Pauli Paths (Fugaku)

<https://www.qedma.com/blog/error-mitigated-quantum-simulation-beyond-classical/>

Quantum Simulation today?

Demonstration	Accuracy evidence	Classical Comparison	Verification
IBM Eagle kicked Ising	Strong for selected observables	Initially incomplete; later surpassed classically	Exact points, mitigation, cross- method checks
Google 69-qubit XY magnet	Strong calibration and XEB; diverse observables	Strong for full chaotic state, less conclusive observable-by- observable	Small-size TN checks, XEB scaling, physical consistency
D-Wave spin- glass dynamics	Good agreement where exact checks exist	Broad but disputed and still evolving	Scaling and observable comparisons
Google Quantum Echoes	Measurable OTOCs with reported signal- to-noise	Strong estimates against known Monte Carlo/tensor	Repetition, hardware/natural- system comparison, learnable parameter
IBM KQD	Systematic convergence but noise-limited	No best-classical win	Variational energy structure and classical subspace solution
Ultracold Floquet Hubbard	Bayesian phase tests, entanglement and high-order correlations	Quantified Frontier estimate and TN failures	Statistical phase checks and lower- scale comparisons

The quantum Simulation problem (informal)



Preliminaries for quantum simulation

Mathematical Preliminaries

1. Hilbert Spaces, Operators and Matrices.
2. Norms and Metrics (distance measures).
3. Big O notation (Asymptotic Notation).
4. Approximation of functions of operators.

Mathematical Preliminaries

1. Hilbert Spaces, Operators and Matrices:

A Hilbert space $\mathcal{H}$ is a complete complex vector space with an inner product defined on the space.

We will usually consider $\mathcal{H} \cong \mathbb{C}^d$

If $|\psi\rangle \in \mathcal{H}$ then

For $|\psi\rangle = (a_1, \dots, a_d)^T$ & $|\phi\rangle = (b_1, \dots, b_d)^T$

$$|\psi\rangle = \begin{pmatrix} a_1 \\ \cdot \\ \cdot \\ \cdot \\ a_d \end{pmatrix} \quad a_1, \dots, a_d \in \mathbb{C}$$

$$\langle\psi|\phi\rangle = \sum_{i=1}^d a_i^* b_i$$

Mathematical Preliminaries

1. Hilbert Spaces, Operators and Matrices:

An operator on the space $\mathcal{H}$ is a linear transformation $A : \mathcal{H} \rightarrow \mathcal{H}$

If $c|\psi\rangle + |\phi\rangle \in \mathcal{H}$ then $A(c|\psi\rangle + |\phi\rangle) = cA|\psi\rangle + A|\phi\rangle \in \mathcal{H}$

Operators can be represented by matrices. If an operator acts on a d -dimensional Hilbert space, then it can be represented by a $d \times d$ matrix.

Mathematical Preliminaries

1. Hilbert Spaces, Operators and Matrices:

One also defines functions of operators. Given a function f and an operator A we can define the function $f(A)$ using the Taylor series,

$$f(A) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (A - x_0 \mathbb{1})^n$$

Of great importance to us will be the exponential function of an operator,

$$\exp(A) = \sum_{n=0}^{\infty} \frac{A^n}{n!} \quad x_0 = 0$$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

A norm is a function defined on the Hilbert space that allows to measure the “length” of a vector in the Hilbert space.

For a Hilbert space $\mathcal{H}$ we can define the norm using the inner product.

For $|\psi\rangle \in \mathcal{H}$

$$\| |\psi\rangle \| = \|\psi\| = \sqrt{\langle \psi | \psi \rangle}$$

Mathematical Preliminaries

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For $|\psi\rangle \in \mathcal{H}$

$$\| |\psi\rangle \| = \|\psi\| = \sqrt{\langle \psi | \psi \rangle}$$

$$\|\psi\| \geq 0 \qquad \|\psi + \phi\| \leq \|\psi\| + \|\phi\| \qquad \|c\psi\| = |c| \|\psi\|, \quad c \in \mathbb{C}$$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

A norm is a function defined on the Hilbert space that allows to measure the “length” of a vector in the Hilbert space.

Examples of vector norms:

p -norm or ℓ^p -norm

$$|x\rangle = (x_1, \dots, x_d)^T \in \mathcal{H} \quad ||x||_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$$

$p = 2$ gives the standard Euclidean norm.

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

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Examples of vector norms:

∞ -norm

$$|x\rangle = (x_1, \dots, x_d)^T \in \mathcal{H}$$

$$\|x\|_\infty := \max_i |x_i|$$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

We can also define a norm for operators

For an operator $A : \mathcal{H} \rightarrow \mathcal{H}$

$$\|A\|_{op} = \sup \left\{ \frac{\|A\psi\|}{\|\psi\|} \mid |\psi\rangle \in \mathcal{H} \right\}$$

The following inequality with the operator norm will be very important,

$$\|A |\psi\rangle\| \leq \|A\|_{op} \|\psi\|$$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

We can also define a norm for matrices a matrix norm that will be used frequently throughout these lectures is the trace norm of a matrix.

Given a $d \times d$ matrix A the trace norm is,

$$\|A\|_{tr} = \text{tr} \left(\sqrt{A^\dagger A} \right)$$

where $A^\dagger = (A^*)^T$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

We can use the norms that we have already mentioned to define a distance measure (a metric) on the Hilbert space and on the space of operators/matrices.

Example:

We can use the p-norm $\|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$ to define a metric for vectors,

$$d_p(|\psi\rangle, |\phi\rangle) = \|\psi - \phi\|_p$$

Mathematical Preliminaries

2. Norms and Metrics (distance measures):

We can use the norms that we have already mentioned to define a distance measure (a metric) on the Hilbert space and on the space of operators/matrices.

Example:

We can use the trace norm $\|A\|_{tr} = \text{tr} \left(\sqrt{A^\dagger A} \right)$ to define a metric for matrices,

$$d_{tr}(A, B) = \frac{1}{2} \|A - B\|_{tr}$$

Mathematical Preliminaries

3. Big O notation (Asymptotic Notation)

Big O notation is a mathematical notation that describes the limiting behaviour of a function when the argument tends towards a particular value or infinity.

Consider a real or complex valued function $f(x)$ and let $g(x)$ be a strictly positive comparison function then,

$$f(x) = O(g(x)) \quad \text{as} \quad x \rightarrow \infty$$

if there exists a positive real number M and a real number x_0 such that

$$|f(x)| \leq M g(x) \quad \text{for all} \quad x \geq x_0$$

Mathematical Preliminaries

3. Big O notation (Asymptotic Notation)

We can also define the Big O notation for the behaviour of $f(x)$ near some value a ,

Consider a real or complex valued function $f(x)$ and let $g(x)$ be a comparison function then,

$$f(x) = O(g(x)) \quad \text{as} \quad x \rightarrow a$$

if there exists a positive real numbers M and a real number x_0 such that

$$|f(x)| \leq M g(x) \quad \text{for all} \quad 0 < |x - a| < x_0$$

Mathematical Preliminaries

3. Big O notation (Asymptotic Notation)

Big O notation is useful when analysing algorithms for efficiency. For example, the time (or the number of steps) it takes to complete a problem of size n might be found to be $T(n) = 4n^2 - 2n + 2$. As n grows large, the term n^2 will come to dominate, so that all other terms can be neglected.

So one will usually write,

$$T(n) = O(n^2)$$

Mathematical Preliminaries

3. Big O notation (Asymptotic Notation)

Big O can also be used to describe the error term in an approximation to a mathematical function. The most significant terms are written explicitly, and then the least-significant terms are summarised in a single big O term.

Consider the Taylor expansion of the exponential function when x is small

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + O(x^3) \quad x \rightarrow 0$$

This means that the absolute value of the error $e^x - 1 - x - \frac{x^2}{2!}$ is at most some constant times $|x^3|$ when $x \rightarrow 0$

Mathematical Preliminaries

4. Approximation of functions of operators:

A large part of quantum simulation is constructing approximations of functions of operators.

Consider an operator A and the operator function $f(A)$ we say that the function $\tilde{f}(A)$ approximates $f(A)$ to a precision $\epsilon > 0$ if

$$\|f(A) - \tilde{f}(A)\|_{op} \leq \epsilon$$

The order of the approximation is the degree of the first non-zero term in the difference of the Taylor expansions of $f(A)$ and $\tilde{f}(A)$

Quantum Mechanics of Closed quantum systems

The state space of a d -dimensional quantum system is the Hilbert space

$$\mathcal{H} \cong \mathbb{C}^d$$

The state of the system at a time $t \geq 0$ is described by a unit vector $|\psi(t)\rangle \in \mathcal{H}$

The dynamics of the system is described by the Schrödinger equation,

$$i \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle$$

where $H = H^\dagger$ is a Hermitian operator called the Hamiltonian.

The Schrödinger equation has the formal solution,

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle = e^{-iHt} |\psi(0)\rangle$$

Quantum Mechanics of Closed quantum systems

$U(t) = e^{-iHt}$ is a unitary operator $U(t)U(t)^\dagger = U(t)^\dagger U(t) = \mathbb{1}$

$$U(t)^\dagger = e^{iHt} = U(-t)$$

Usually one is given the Hamiltonian in the form

$$H = \sum_j H_j$$

Therefore if one wants to describe the dynamics of the system one must be able to realise

$$U(t) = e^{-i \sum_j H_j t}$$

Why is it hard to simulate quantum dynamics on a classical computer?

We can always represent $U(t)$ as a $d \times d$ matrix.

Consider a system of n spins. The Hamiltonian describing a single spin is a 2×2 matrix.

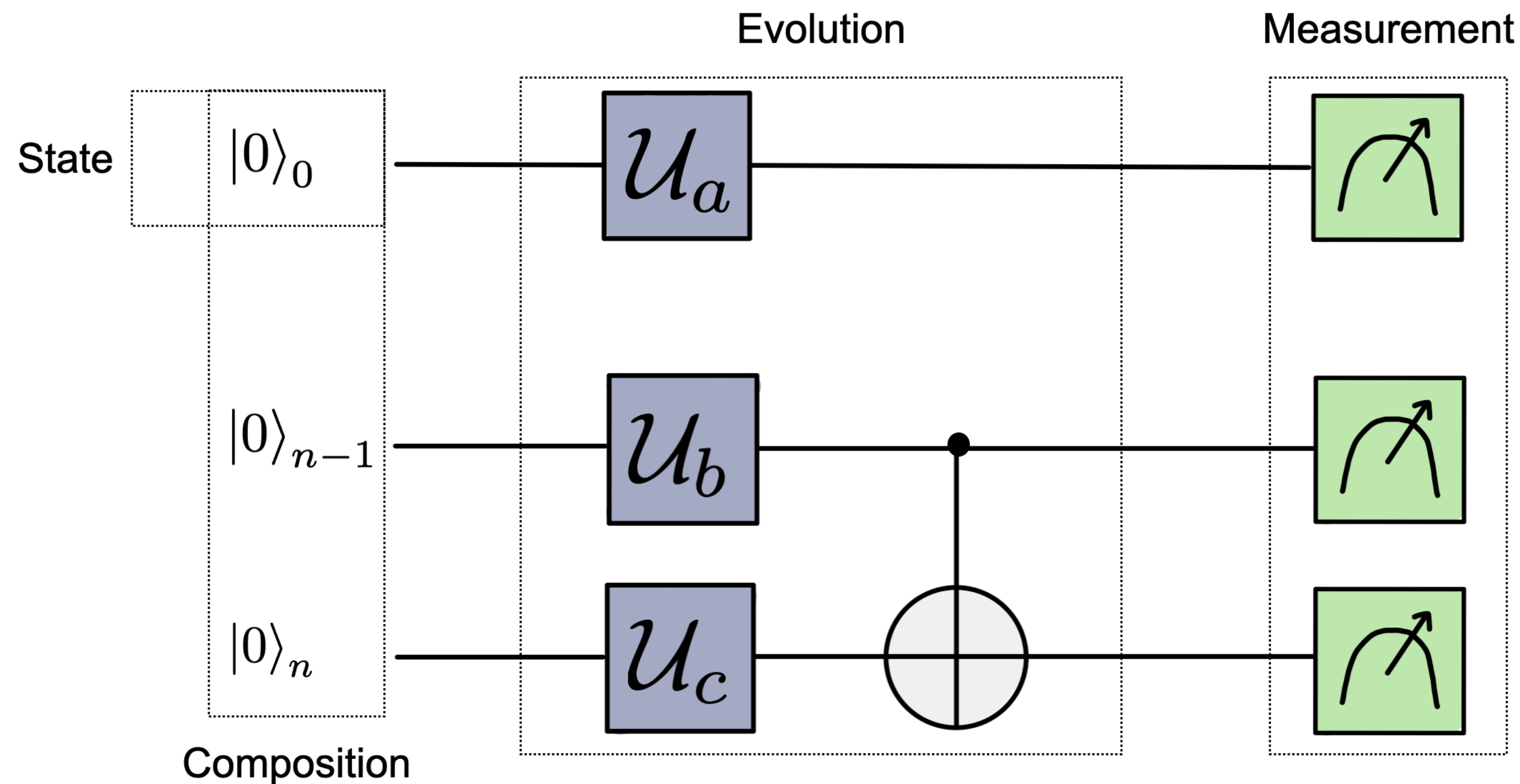
So the Hamiltonian of n spins is a $2^n \times 2^n$ matrix.

This means that the unitary operator describing the evolution will also be a $2^n \times 2^n$ matrix.

So the size of the matrix that a classical computer needs to store will grow exponentially with the number of spins.

Basic Quantum computing

In quantum computing we make use of quantum systems called qubits and we perform computations by applying a sequence of quantum gates (unitary operations) to these qubits and then we perform measurements to obtain the output the computation.



The quantum Simulation problem

Given an initial state $|\psi(0)\rangle$, a simulation time $t \geq 0$, a precision $\epsilon > 0$ and a Hamiltonian $H = \sum_j H_j$

Construct a unitary approximation $\tilde{U}(t)$ to the unitary operator $U(t) = e^{-i \sum_j H_j t}$ such that

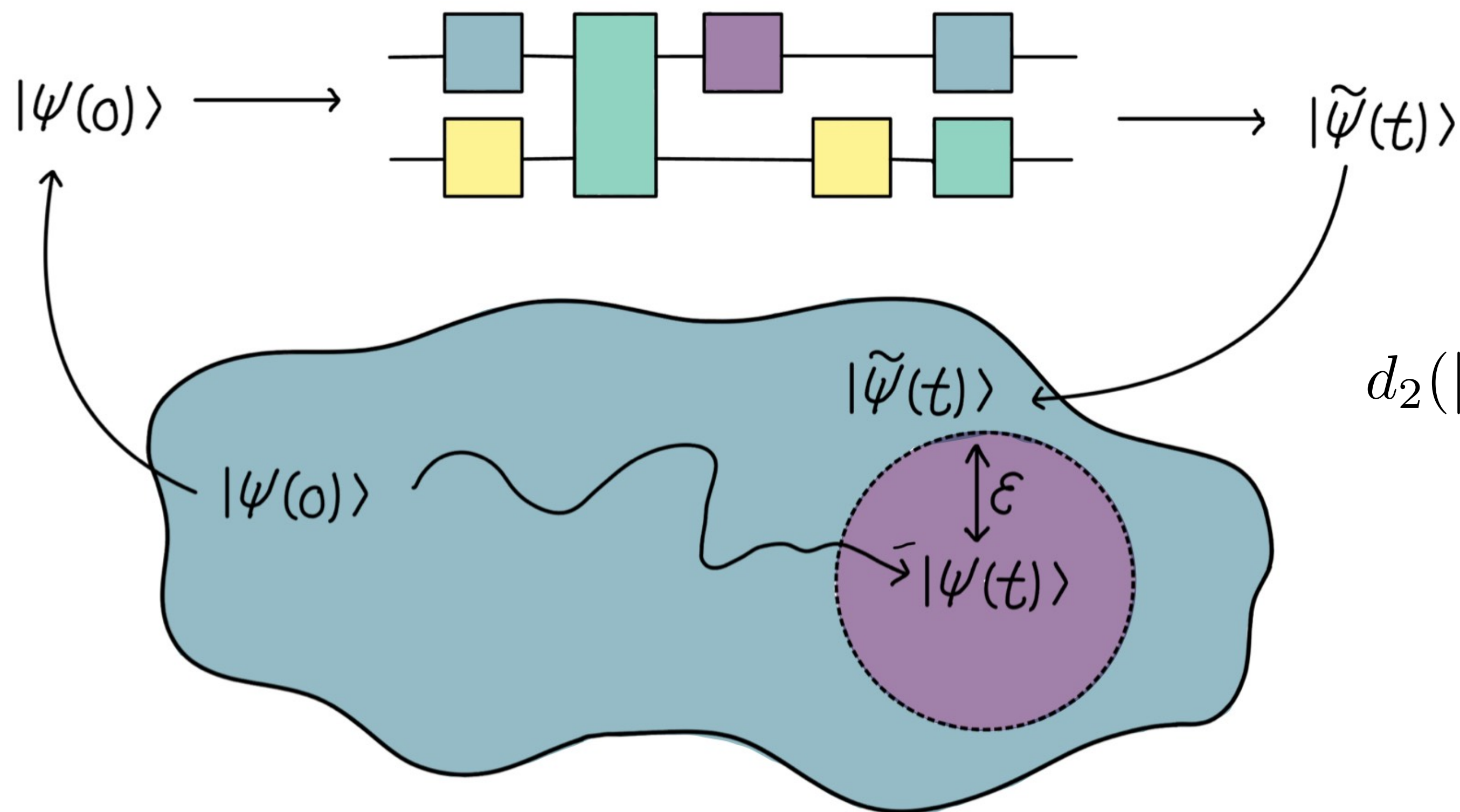
$$\|U(t) - \tilde{U}(t)\|_{op} \leq \epsilon$$

and the approximation $\tilde{U}(t)$ can be implemented efficiently with a polynomial number of quantum gates.

The quantum Simulation problem

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

$$|\tilde{\psi}(t)\rangle = \tilde{U}(t) |\psi(0)\rangle$$



$$\begin{aligned} d_2(|\psi(t)\rangle, |\tilde{\psi}(t)\rangle) &= \| |\psi(t)\rangle - |\tilde{\psi}(t)\rangle \|_2, \\ &\leq \|U(t) - \tilde{U}(t)\|_{op} \| |\psi(0)\rangle \|_2, \\ &\leq \|U(t) - \tilde{U}(t)\|_{op} \leq \epsilon \end{aligned}$$

First order Trotter-Suzuki product formulas

Let us try to do this for a simple Hamiltonian we want to simulate,

$$H = H_1 + H_2$$

This means we need to construct an approximation to,

$$U(t) = e^{-i(H_1+H_2)t}$$

as a product of other unitaries. The natural thing to do is to try to write $\tilde{U}(t)$ in terms of e^{-iH_1t} and e^{-iH_2t} , to do this we recall the Trotter Formula

$$e^{-i(H_1+H_2)t} = \lim_{n \rightarrow \infty} (e^{-iH_1t/n} e^{-iH_2t/n})^n.$$

The Trotter formula is an exact formula if we want to approximate $U(t)$ then we know that for large enough n we have,

$$e^{-i(H_1+H_2)t} \approx (e^{-iH_1t/n}e^{-iH_2t/n})^n.$$

However to use this approximation we need a rigorous bound for the error as well as a bound on n as saying large enough isn't precise enough.

Let us start by defining $\tau := \frac{t}{n}$ then we can write,

$$U(t) = \left(e^{-iHt/n} \right)^n = \left(e^{-iH\tau} \right)^n = U(\tau)^n$$

and,

$$\tilde{U}(\tau) = e^{-iH_1\tau} e^{-iH_2\tau}.$$

Then we need to show that,

$$\left\| U(t) - \tilde{U}(\tau)^n \right\| \leq \epsilon$$

and we need to find lower bounds on the precision $\epsilon > 0$ and on $n \in \mathbb{N}$.

Using the identity $\|A^p - B^p\| \leq p\|A - B\|$ we can write,

$$\|U(t) - \tilde{U}(\tau)^n\| = \|U(\tau)^n - \tilde{U}(\tau)^n\| \leq n\|U(\tau) - \tilde{U}(\tau)\|$$

Therefore if we want complete the proof we need to bound $\|U(\tau) - \tilde{U}(\tau)\|$.

To do this we need to make use of the Taylor expansion. We start by expanding $U(\tau)$,

$$U(\tau) = e^{-iH\tau} = \sum_{k=0}^{\infty} \frac{(-iH\tau)^k}{k!} = \sum_{k=0}^{\infty} \frac{(-i\tau)^k}{k!} (H_1 + H_2)^k.$$

Next we need to Taylor expand each of the exponentials in $\tilde{U}(\tau)$,

$$\tilde{U}(\tau) = e^{-iH_1\tau} e^{-iH_2\tau} = \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l.$$

If we take the difference here we see that all terms that have τ^0 and τ^1 cancel leaving the following remainder terms,

$$U(\tau) - \tilde{U}(\tau) = \sum_{k=2}^{\infty} \frac{(-i\tau)^k}{k!} H^k - \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l$$

We can now take the norm of both sides,

$$\begin{aligned} \|U(\tau) - \tilde{U}(\tau)\| &= \left\| \sum_{k=2}^{\infty} \frac{(-i\tau)^k}{k!} H^k - \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l \right\| \\ &\leq \left\| \sum_{k=2}^{\infty} \frac{(-i\tau)^k}{k!} H^k \right\| + \left\| \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l \right\| \end{aligned}$$

where we used the triangle inequality $\|A + B\| \leq \|A\| + \|B\|$.

Consider the first term,

$$\left\| \sum_{k=2}^{\infty} \frac{(-i\tau)^k}{k!} H^k \right\| \leq \sum_{k=2}^{\infty} \frac{|-i\tau|^k}{k!} \|H^k\| \leq \sum_{k=2}^{\infty} \frac{|-i\tau|^k}{k!} \|H\|^k$$

where we have once again used the triangle inequality and the sub-multiplicative property $\|A^k\| = \|A\|^k$ for $k \in \mathbb{Z}$.

To finish this bound we need a bound on the norm of the Hamiltonian. If we define $\Lambda := \max\{\|H_1\|, \|H_2\|\}$ then,

$$\|H\| \leq \|H_1\| + \|H_2\| \leq 2\Lambda.$$

This leads us to the following bound,

$$\left\| \sum_{k=2}^{\infty} \frac{(-i\tau)^k}{k!} H^k \right\| \leq \sum_{k=2}^{\infty} \frac{|-i\tau|^k (2\Lambda)^k}{k!}.$$

Looking at the second term we have,

$$\left\| \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l \right\| \leq \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{|-i\tau|^{k+l}}{k! l!} \|H_1\|^k \|H_2\|^l$$

Now we observe that,

$$\|H_1\| \leq \Lambda \quad \text{and} \quad \|H_2\| \leq \Lambda$$

Therefore,

$$\left\| \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l \right\| \leq \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{|-i\tau|^{k+l} \Lambda^{k+l}}{k! l!}$$

Now we rewrite the two infinite sums here as a restricted sum,

$$\sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{|-i\tau|^{k+l} \Lambda^{k+l}}{k! l!} = \sum_{p=2}^{\infty} \sum_{\substack{k,l=0 \\ k+l=p}}^p \frac{|-i\tau|^{k+l} \Lambda^{k+l}}{k! l!}$$

All that remains is to find a way to compute the sum. Consider the following,

$$e^{2x} = e^x e^x$$

we can expand both sides in a Taylor series as,

$$\sum_{p=0}^{\infty} \frac{(2x)^p}{p!} = \sum_{q=0}^{\infty} \sum_{s=0}^{\infty} \frac{x^{q+s}}{q! s!} = \sum_{p=0}^{\infty} \sum_{\substack{q,s=0 \\ q+s=p}}^p \frac{x^{q+s}}{q! s!}$$

This immediately tells us that,

$$\frac{(2x)^p}{p!} = \sum_{\substack{q,s=0 \\ q+s=p}}^p \frac{x^{q+s}}{q! s!}$$

Using this result with $x = | - i\tau|\Lambda$ we have,

$$\left| \left| \sum_{k=2}^{\infty} \sum_{l=2}^{\infty} \frac{(-i\tau)^{k+l}}{k! l!} H_1^k H_2^l \right| \right| \leq \sum_{p=2}^{\infty} \frac{| - i\tau|^p (2\Lambda)^p}{p!}$$

Putting it all together we have,

$$\begin{aligned}\left\|U(\tau) - \tilde{U}(\tau)\right\| &\leq \sum_{k=2}^{\infty} \frac{|-i\tau|^k (2\Lambda)^k}{k!} + \sum_{p=2}^{\infty} \frac{|-i\tau|^p (2\Lambda)^p}{p!} \\ &= 2 \sum_{p=2}^{\infty} \frac{|-i\tau|^p (2\Lambda)^p}{p!}\end{aligned}$$

If we recall the following identity for the bound of the exponential function,

$$\sum_{p=k}^{\infty} \frac{|\lambda|^p}{p!} \leq \frac{|\lambda|^k}{k!} e^{|\lambda|}$$

This allows us to arrive at the bound,

$$\left\| U(\tau) - \tilde{U}(\tau) \right\| \leq 2 \frac{|-i\tau|^2 (2\Lambda)^2}{2!} e^{2|-i\tau|\Lambda} = (2\tau\Lambda)^2 e^{2\tau\Lambda}$$

Since $\tau = t/n$ we have,

$$\left\| U(t) - \tilde{U}(\tau)^n \right\| \leq n \frac{(2t\Lambda)^2}{n^2} e^{2t\Lambda/n} = \frac{(2t\Lambda)^2}{n} e^{2t\Lambda/n}$$

If we choose $\epsilon \geq \frac{(2t\Lambda)^2}{n} e^{2t\Lambda/n}$ then $\left\| U(t) - \tilde{U}(\tau)^n \right\| \leq \epsilon$.

With this rigorous bound on the accuracy to which $\tilde{U}$ approximates U we usually say that,

$$U(\tau) = \tilde{U}(\tau) + O(\tau^2)$$

which means that $\tilde{U}$ is a first order approximation as the error term is quadratic.

In a quantum simulation problem we usually want to specify a precision ϵ and then find the number of times n we need to apply $\tilde{U}(\tau)$.

To find n one either directly solves numerically $\epsilon = \frac{(2t\Lambda)^2}{n} e^{2t\Lambda/n}$, or if we choose some n such that $e^{2t\Lambda/n} \approx 1$ then we can easily have,

$$n = \left\lceil \frac{(2t\Lambda)^2}{\epsilon} \right\rceil.$$

More generally we have that $H = \sum_{j=1}^L H_j$ so using the same method as we did above we construct $\tilde{U}(\tau)$,

$$\tilde{U}(\tau) := \prod_{j=1}^L e^{-iH_j\tau},$$

we then show that,

$$\left\| U(t) - \tilde{U}(\tau)^n \right\| \leq \frac{(Lt\Lambda)^2}{n} e^{Lt\Lambda/n} \leq \epsilon$$

where $\Lambda := \max_j \|H_j\|$. Here again we see that $\tilde{U}$ approximates U up to first order. We find that n should be,

$$n \geq \left\lceil \frac{(Lt\Lambda)^2}{\epsilon} \right\rceil.$$

Higher order Trotter-Suzuki product formulas

So far we have looked at an approximation up to only first order but what about higher order approximations. This problem was addressed by Suzuki

To keep track of the order to which we approximate U we will write down our product approximations as S_k where k will denote the order to of the approximation, for example

$$\tilde{U}(\tau) = S_1(\tau) = \prod_{j=1}^M e^{-iH_j\tau}.$$

Suzuki M. Fractal decomposition of exponential operators with applications to many-body theories and Monte Carlo simulations. Physics Letters A. 1990 Jun 4;146(6):319-23.

As a starting point consider the second order approximation, which for a Hamiltonian $H = \sum_{j=1}^M H_j$ and a small time step $\tau \geq 0$ is,

$$S_2(\tau) = \prod_{j=1}^M e^{-iH_j\tau/2} \prod_{j'=M}^1 e^{-iH_{j'}\tau/2}$$

The error bound for this formula is, $\|U(t) - S_2(\tau)^n\| \leq \frac{(2L\tau\Lambda)^3}{3n^2} e^{2L\tau\Lambda/n}$, which shows that it is a second order formula and a value for n is found in a similar manner to the first order formula.

We also note that this approximation is symmetric meaning that under time reversal the structure is preserved.

Suzuki started with the second order product formula S_2 and recursively constructed higher order product formulas.

He did this by taking products of the second order product formulas in such a way that when you Taylor expand it you are able to cancel higher order terms in the expansion of $U(t)$.

As an example, Suzuki found that if you consider the Taylor expansion of the product,

$$S_{2k-2}(p_2\tau)^2 S_{2k-2}([1 - 4p_2]\tau) S_{2k-2}(p_k\tau)^2$$

where $p_k = 1/(4 - 4^{(1/2k-1)})$, then this product formula can approximate $U(\tau)$ to an order $2k$ where $k \in \mathbb{N}$. We will usually write this as $S_{2k}(\tau)$ i.e.

$$S_{2k}(\tau) = S_{2k-2}(p_2\tau)^2 S_{2k-2}([1 - 4p_2]\tau) S_{2k-2}(p_k\tau)^2.$$

Any formula of this form is referred to as a Trotter-Suzuki product formula.

The analysis for the error bounds with this product formula are quite complicated so we just state the bounds,

$$\|U(t) - S_{2k}(\tau)^n\| \leq \frac{(2L5^{k-1}\Lambda t)^{2k+1}}{3n^{2k}} \exp\left(\frac{2L5^{k-1}\Lambda t}{n}\right) \leq \epsilon$$

and we find n to be,

$$n = \left\lceil \max \left\{ 2L5^{k-1}\Lambda t, \sqrt[2k]{\frac{e(2L5^{k-1}\Lambda t)^{2k+1}}{3\epsilon}} \right\} \right\rceil.$$

An important quantity that will aid us later in computing the number of gates in our circuit is the number of exponentials in the product formula N_{exp} which for this formula is,

$$N_{exp} \leq 2L5^{k-1}n.$$

**We can now write down our first quantum simulation
algorithm!!**

Trotter-Suzuki Quantum simulation Algorithm

Step 1: Given the Hamiltonian H for our system write it in the form $H = \sum_j H_j$ such that each H_j can be exponentiated easily and has an efficient quantum circuit implementation.

Step 2: Pick an order k for the Trotter-Suzuki product formula S_{2k} you wish to use to approximate $U(t)$.

Step 3: Compute $\Lambda = \max_j \|H_j\|$, specify a simulation time $t \geq 0$, a precision $\epsilon > 0$ and some initial state $|\psi(0)\rangle$. Use this to compute n .

Step 4: Construct quantum circuits for each $e^{-iH_j\tau}$.

Step 5: Construct a gate set using $S_{2k}(t/n)^n$ that will be implemented on a quantum computer using $|\psi(0)\rangle$ as an initial state.

Example: Simulating the XX spin chain

Let us consider the Hamiltonian for the XX spin chain,

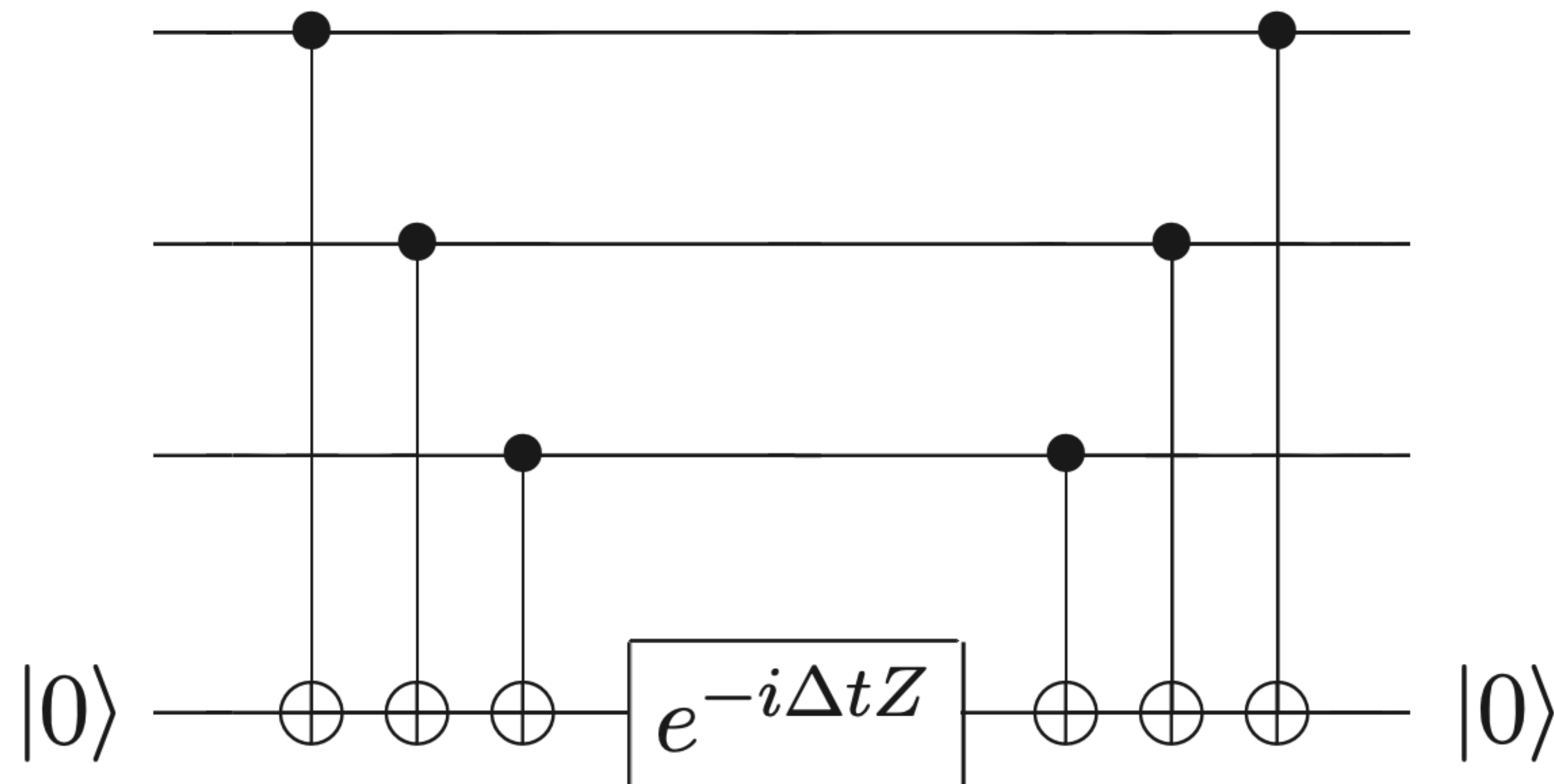
$$H = \sum_{j=1}^{M-1} \sigma_x^{(j)} \sigma_x^{(j+1)} + \sigma_y^{(j)} \sigma_y^{(j+1)}$$

where $\sigma_x^{(j)}$ is just σ_x applied to the j -th spin and for our example we choose $M = 3$ so that the spin chain has 3 sites. If we want to simulate this Hamiltonian we need to write H in a form such that we can exponentiate H_j easily.

$$H_1 = \sigma_x^{(1)} \sigma_x^{(2)}, \quad H_2 = \sigma_x^{(2)} \sigma_x^{(3)}, \quad H_3 = \sigma_y^{(1)} \sigma_y^{(2)}, \quad H_4 = \sigma_y^{(2)} \sigma_y^{(3)}$$

and $H = \sum_{j=1}^4 H_j$.

The reason that we choose this form for H is because circuits for $e^{-iH_j\tau}$ can be constructed easily. We note that we can construct an efficient circuit to simulate $e^{-i(Z\otimes Z\otimes Z)\tau}$ using just one ancilla qubit i.e.



Now we recall that we can transform between σ_x, σ_y and σ_z using single qubit unitaries. So if one wants to simulate $e^{-i(Z \otimes X \otimes Z)\tau}$ we can use the relation,

$$U e^{-iH\tau} U^\dagger = e^{-iU H U^\dagger \tau}$$

so that we can simulate,

$$(\mathbb{1} \otimes \hat{H} \otimes \mathbb{1}) e^{-i(Z \otimes Z \otimes Z)\tau} (\mathbb{1} \otimes \hat{H} \otimes \mathbb{1}) = e^{-i(Z \otimes X \otimes Z)\tau}$$

$\hat{H}$ is the Hadamard gate.

Exercise: Find quantum circuits for each $e^{-iH_j\tau}$ using the relations above.

Now that we have written our Hamiltonian in the correct form we need to choose a Trotter-Suzuki formula for our approximation in most practical cases one chooses $k = 2, 4$.

We specify a simulation time $t = 2\text{s}$ and the most common precision that is chosen in literature is $\epsilon = 10^{-6}$. For our Hamiltonian we find $\Lambda = 1$.

If we choose $k = 2$ then we have that

$$n = \lceil (8)^{3/2} / \sqrt{3 \times 10^{-6}} \rceil = 13\,064$$

and the number of exponentials will be,

$$N_{exp} = 2 \lceil (8)^{3/2} / \sqrt{3 \times 10^{-6}} \rceil = 26\,128$$

Post Trotter methods:

1. Hamiltonian Simulation By Truncated Taylor Series

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2. Hamiltonian simulation By Quantum Signal Processing and Qubitization.

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Low GH, Yoder TJ, Chuang IL. Methodology of resonant equiangular composite quantum gates. Physical Review X. 2016 Oct 1;6(4):041067.



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